

Thesis Chapter Outline

Chapters 2 to 6 — structure, visuals and the claim to make in each section

Two structural decisions are built into this outline. (i) The modified M2M ratio is promoted to its own chapter, placed immediately after Chapter 1 so that it follows directly from §1.6, where the conventional ratio is introduced. (ii) Chapters 2 and 3 are written entirely in generic block notation (M, X, Y) and name no process, so both case studies inherit them unchanged. The case studies are introduced for the first time in Chapter 4.

	Chapter	Role	Length
1	Mathematical background	unchanged; ends at §1.6 the M2M ratio	~2,900
2	A Modified M2M Ratio	NEW — the contribution	~3,200
3	Methodology	generic procedure	~2,400
4	Investigated Processes and Data Generation	both case studies	~3,000
5	Results and Discussion	all findings	~4,500
6	Conclusions	contribution, limits, future work	~1,200

Chapter 2 — A Modified M2M Ratio

Target length ~3,200 words. Case studies are NOT named in this chapter.

2.1 Limitations of the conventional ratio

Open from §1.6. The conventional ratio is formed from a sequential decomposition in which the mechanistic block enters first, so the variance that both blocks could explain is credited entirely to the numerator. State the two consequences: the ratio grows whenever the blocks overlap, independently of how good the mechanism is; and it carries no reference value, so a single number cannot be interpreted on its own.

CLAIM A metric usable only by comparison cannot certify one model — which is what an engineer deciding whether to trust a mechanism needs.

TO MAKE Fig. 2.1 — set diagram of $\text{joint} = u_M + u_X + \text{shared}$, with the conventional numerator shaded to show it absorbing the shared region. One generic schematic, no data.

2.2 Commonality decomposition of the block contributions

Define the four quantities by equation and in words. State that they are obtained by refitting the model with one block removed and differencing, and that when a prior block is present the decomposition is conditional on it. This section absorbs the block-level half of the current §2.3; the species-level half moves to Chapter 3.

TO MAKE Table 2.1 — the four quantities: symbol, definition as a Q^2 difference, and plain-language interpretation.

2.3 Definition of the modified ratio and its intrinsic reference

Give $M2M^* = u_M/u_X$. State precisely what the reference at 1 means: it is the point at which the two blocks supply equal unique information, so the threshold is a property of the decomposition rather than a convention. Say what a value above 1 does and does not license.

CLAIM The reference is intrinsic, not calibrated: no second model and no tuning constant is required to interpret a single value.

2.4 Why the shared term is excluded

Three independent arguments, in this order.

(i) **Structural.** shared is unchanged when M and X are swapped. A quantity that ranks one block against the other must be antisymmetric under that swap; the ratio inverts, shared does not. It is therefore incapable in principle of expressing the comparison.

(ii) **Informational.** When the joint explained variance and u_M are stable across candidates, shared reduces to a constant minus u_X — so it carries nothing the ratio does not already use, and uses it without an orientation. Forward-reference the demonstration to §5.4.

(iii) **Decision-theoretic.** The advantage of the hybrid over the better single-source model is $\min(u_M, u_X)$, which is bounded by $(\text{joint} - \text{shared})/2$. A large shared term therefore argues *against* hybridising, not for it.

CLAIM Dropping the shared term costs no ranking information and buys the reference point. It is reported alongside the ratio, never inside it.

2.5 Estimation: cross-validated Q^2 rather than sums of squares

In-sample sequential sums of squares increase monotonically with latent variables, so a ratio built from them drifts with the allocation. Cross-validated Q^2 penalises added components and pins the ratio down. State that u_M and u_X are therefore out-of-sample gains and that the metric cannot be inflated by adding components.

Also fix the **sub-model protocol**: each reduced model is re-optimised on its own allocation grid rather than inheriting the full model's. Justify it — each sub-model should be given its best chance, or the difference is not a fair measure of what the removed block added.

CLAIM Forward-reference §5.6: under the sums-of-squares estimator the ratio for a correct mechanism crosses the reference value within the set of statistically equivalent allocations, so the threshold does not survive. Under cross-validation it does.

2.6 Latent-variable allocation and the stability of the ratio

The optimal allocation is not unique. Refer forward to §3.3 for the 1-SE flat region and state the consequence here: the metric should be reported over that region rather than at a single point. Give the variance propagation $\text{var}(M2M^*) \approx (1/u_X^2) \cdot [\text{var}(u_M) + M2M^{*2} \cdot \text{var}(u_X)]$ and draw the conclusion that precision is controlled by the denominator.

Introduce averaging over repeated cross-validation partitions, and report $P(M2M^* > 1)$ across them as the primary summary — it is the quantity the threshold actually licenses, and it is far more stable than the magnitude.

2.7 Properties and scope conditions

State the limits openly; this section protects the contribution rather than weakening it.

Scale-freedom. The ratio depends only on the proportion of the uniques, so $u_M/u_X = 0.05/0.05$ and $0.40/0.40$ give the same value while describing very different models. Define coverage $= (u_M + u_X)/\text{joint}$ as the companion statistic.

CAUTION State explicitly that coverage is **not** a quality criterion and must not be maximised jointly with the ratio — the two oppose each other, because a good mechanism drives u_X down. Evidence in §5.10.

Degeneracy. $u_X \rightarrow 0$ when the mechanism can forecast the window in which the response is measured; the ratio then diverges and only its side of 1 remains meaningful.

Design requirements. Sample size relative to block width, and the requirement that the response lie outside the window spanned by both blocks.

TO MAKE Table 2.2 — applicability checklist: condition, why it matters, what to report when it fails.

Chapter 3 — Methodology

Target length ~2,400 words. Case studies are NOT named in this chapter.

Everything here is stated for an arbitrary set of candidate mechanistic models and an arbitrary process. No case study is named. Both case studies then reference this chapter without restating it.

3.1 Data preprocessing

Keep the current §2.1 essentially as written — autoscaling, block scaling by $1/\sqrt{K}$ for MB-PLS, and train-fold-only scaling to prevent leakage. Add one sentence on how near-constant columns are handled, since mechanistic blocks can contain them.

3.2 Implementation of the multiblock models

Keep the current §2.2, with the three algorithm boxes and the comparison table. This is the section to expand if Chapter 3 reads thin.

CAUTION Fix the inconsistency: the current text says latent variables were chosen by minimising RMSECV. Change to the parsimonious member of the 1-SE region and cross-reference §3.3, or the chapter contradicts itself.

TABLE **Table 3.1** — comparison of MB-PLS, SO-PLS and SMB-PLS (exists).

3.3 Latent-variable selection and the flat region

Current §2.5. Define the grid search, the Måge plot, the 1-SE threshold and the parsimonious selection rule within the region. State the rule once, precisely, since Chapters 2 and 5 both depend on it.

TO MAKE **Fig. 3.1** — generic Måge-plot schematic: RMSECV against total components, flat band marked, argmin and parsimonious choices labelled. Draw it with unlabelled axes so it stays case-study free.

3.4 Species-level contribution analysis

The species-removal half of the current §2.3. Keep it here: it is a diagnostic procedure applied to the models, not part of the definition of the metric.

VISUAL **Fig. 3.2** — species-level removal scheme (exists as current Figure 2.1).

3.5 Extrapolation validation

Current §2.6. Describe the rolling scheme generically: at decision point k the measured trajectory up to k is combined with knowledge-driven forecasts beyond it, and the end-of-batch response is predicted. Define the performance measure and the rank-correlation comparison against the diagnostics.

VISUAL **Fig. 3.3** — extrapolation procedure schematic (exists as current Figure 2.3).

CAUTION State here that the diagnostic is computed on calibration data only and the extrapolation set is never used to construct it. This is the sentence that makes the validation in §5.8 admissible.

3.6 Software and reproducibility

Short. Custom implementations in Python with NumPy, SciPy and scikit-learn; fixed cross-validation seeds; each case study reproducible from a single script. One paragraph, no figures.

Chapter 4 — Investigated Processes and Data Generation

Target length ~3,000 words.

4.1 Selection of the case studies

State the design logic before either process appears: one case study with a family of candidate mechanisms of graded, known quality, to test whether the metric ranks as an oracle would; and one taken from independent published work, to test it on a misspecification the literature itself considers realistic rather than one constructed here.

4.2 Two-stage batch reactor

4.2.1 Process description — the two stages, the desired and impurity routes, the response. 4.2.2 Data generation — random operating conditions, the stochastic differential equation formulation, measurement noise, number of batches. 4.2.3 Candidate model variants. 4.2.4 Data blocks.

VISUAL Fig. 4.1 — reaction scheme for both stages.

TO MAKE Table 4.1 — the seven candidates: what was altered in each, the resulting rate laws, and the independent kinetic fit cost. Colour or group by whether the reaction network survives.

CLAIM The candidates span parameter-level errors, rate-law errors and structural deletions, and each carries an independent quality grade, so the metric can be checked against an oracle rather than against intuition.

TABLE Table 4.2 — data blocks: X■ 100×6, M■ 100×45, X■ 100×42, Y 100×1 (exists in the current draft).

4.3 Urethane manufacturing process

4.3.1 Process description and the reversible step. 4.3.2 Provenance: what was taken from the original authors and what had to be reconstructed, stated explicitly. 4.3.3 Validation of the reconstruction. 4.3.4 The two candidate models defined by the source papers. 4.3.5 Data blocks.

VISUAL Fig. 4.2 — reaction scheme. Fig. 4.3 — published trajectories against the reconstruction (cs2_compare_published.png).

CLAIM Reconstruction validated at 5.0 % RMS deviation across six panels; and the authors' individual noise realisation is reproduced, not merely its variance — correlation 0.60 against -0.005 ± 0.055 for 25 alternative seeds, $z = 11.1$.

TABLE Table 4.3 — data blocks: M 40×906, X 40×423, Y 40×1 (deck_cs2_blocks_table.png).

CAUTION State plainly that the authors' input workbook was unavailable, so the batches are sampled around one validated run rather than being their experimental design. This is a reconstruction, not a replication.

4.4 Comparison of the two designs

A short closing table contrasting the two: number of blocks, presence of a prior block, sample-to-variable ratio, source of the misspecification, and whether an independent quality grade exists. This sets up which results in Chapter 5 are expected to be weaker and why.

TO MAKE Table 4.4 — design comparison. Flag the 2:1 versus 1:23 sample-to-variable contrast here; §5.10 returns to it.

Chapter 5 — Results and Discussion

Target length ~4,500 words.

Order is deliberate: establish that the problem is real (5.1), show the decomposition (5.2), show the metric works (5.3), explain *why* it works (5.4), show it is stable (5.5-5.6), compare with the conventional ratio (5.7), validate externally (5.8), replicate independently (5.9), and state the limits (5.10).

5.1 In-domain performance does not separate the candidates

Open with the motivating negative result. Report the predictive performance of every candidate and show that it is effectively flat.

CLAIM All seven candidates reach $Q^2 = 87.8 - 90.4\%$, a span of 2.5 percentage points, while differing by more than 17 R^2 units out of domain. In-domain fit cannot rank mechanisms — which is the entire reason a contribution metric is needed.

SOURCE `check_lv_rule.xlsx`, `case_study_1_rolling.xlsx`

5.2 Commonality decomposition of the candidates

Present the full decomposition before any ratio is formed, so the reader sees the raw quantities first.

VISUAL **Fig. 5.1** — stacked commonality bars per candidate (`rep_cs1_commonality.png`).

TABLE **Table 5.1** — Q^2 full, Q^2 without M, Q^2 without X, u_M , u_X , shared, coverage for each candidate.

CLAIM The shared term dominates throughout (56 – 80 % of the jointly explained variance), so coverage is 0.20 – 0.44 and the ratio arbitrates a minority of the model. Say this here rather than letting a reader discover it.

5.3 The modified ratio separates sound from broken mechanisms

The headline result. Report the ratio with its dispersion across repeated cross-validation partitions and the probability of exceeding the reference.

VISUAL **Fig. 5.2** — ratio per candidate with error bars and the reference line at 1, coloured by whether the mechanism is sound (`deck_cs1_ratio.png`).

TABLE **Table 5.2** — the seven-candidate table: Q^2 full, Q^2 no-M, Q^2 no-X, u_M , u_X , $M2M^*$, $\pm$ sd, $P(>1)$. (`check_lv_rule.xlsx`)

CLAIM Under the parsimonious 1-SE rule the four sound mechanisms score 1.67 – 2.03 with $P(>1) = 1.00$, and the three broken ones 0.36 – 1.08. Report the argmin column too (1.96 – 2.11 against 0.38 – 1.16) and state that the conclusion does not depend on the selection rule.

CAUTION Be explicit that one broken candidate sits just above the reference (1.08, $P(>1) = 0.75$). Do not round it away — it is the honest boundary case and it is better volunteered than found.

5.4 Where the discriminating power comes from

This section converts the result from an observation into an explanation, and it is what makes the contribution defensible rather than empirical.

VISUAL **Fig. 5.3** — u_M and u_X side by side per candidate (`deck_cs1_uniques.png`).

CLAIM u_M is nearly constant across candidates (10 – 13 %); it is u_X that moves, from about 6 % when the mechanism is sound to 16 – 27 % when it is not. The metric detects a failing mechanism by observing that the measurements are forced to compensate.

CLAIM Tie this to the construction in §4.2.3: the threshold falls exactly where the reaction *network* breaks, not where parameters are merely wrong. Candidates that keep both reaction channels score near 2; those that delete a channel or remove a dependence fall to 1 or below.

CLAIM Deliver the empirical half of §2.4 here: shared and u_X are rank-equivalent (Spearman = -1.000) when u_M is stable, and the equivalence breaks (-0.286) when u_M varies nine times more. Shared is u_X without an orientation, which is why it is not used.

5.5 Stability across statistically equivalent allocations

Show that the result does not depend on which member of the flat region is chosen.

VISUAL **Fig. 5.4** — Måge plot: full RMSECV landscape with the flat region marked, beside the same region coloured by the ratio, with the argmin and parsimonious allocations labelled (`deck_m0_flat_region.png`).

CLAIM Across all 233 allocations inside the 1-SE band the ratio spans 1.59 – 1.82, a 14 % band, and the two candidate selections differ by 4 % (1.60 against 1.67). The allocation choice does not change the conclusion.

CLAIM Note that the sub-model allocation matters more than the full-model allocation (about ± 0.5 against ± 0.1). This justifies fixing the sub-model protocol in §2.5 and is a result in its own right.

VISUAL **Fig. 5.5** — Q^2 cost against parsimony gained for each stopping rule (`rep_cs1_q2_cost.png`).

CLAIM Averaged over the candidates, the first standard error removes 3.7 latent variables for 0.014 of Q^2 ; the second costs a further 0.019 to remove only 1.6 more. Parsimony is nearly exhausted after one standard error while the predictive cost accelerates — the justification for stopping at 1-SE.

5.6 Estimator choice: cross-validated Q^2 against sums of squares

The methodological result that justifies §2.5. Present it as a comparison, not as an assertion.

VISUAL **Fig. 5.6** — the same flat region coloured by the sums-of-squares metrics, three panels (`deck_m0_flat_region_ss.png`).

TABLE **Table 5.3** — the seven candidates under the sums-of-squares estimator (`table_ss_estimator.xlsx`).

CLAIM The sums-of-squares estimator ranks correctly but loses the threshold: a candidate with an entire reaction deleted scores 1.49 and 1.02, above the reference, on both selection rules. Under cross-validation the same candidate scores 0.70 and 0.67.

CLAIM Worse, six of the seven candidates have flat regions that straddle the reference, including all four sound ones: roughly 10 % of statistically equivalent allocations would declare a sound mechanism data-driven. Spread 0.69 – 6.80 against 1.59 – 1.82 for cross-validation.

CLAIM Conclude: the modified ratio requires cross-validated Q^2 . In-sample sums of squares reward components monotonically, so the ratio drifts with the allocation and the reference washes out.

5.7 Conventional against modified ratio

Direct comparison on the same candidates, same allocations.

VISUAL **Fig. 5.7** — the two ratios side by side with the reference line (`deck_cs1_vs.png`).

CLAIM Both order the candidates correctly. The conventional ratio places every candidate above 1 (2.4 – 12.6), so it yields an ordering and no verdict; the modified ratio places sound and broken mechanisms on opposite sides of a threshold that was not fitted to the data.

CAUTION Do not overclaim here. On ranking alone the conventional ratio is not worse. The contribution is interpretability, and saying so plainly is stronger than implying superiority on every axis.

5.8 External validation against out-of-domain performance

The section that gives the metric something outside itself to be right about.

VISUAL **Fig. 5.8** — rolling prediction error against decision point, within domain and extrapolated (`rep_cs1_rolling.png`).

Fig. 5.9 — the ratio against kinetic fit cost and against extrapolation performance (`rep_cs1_metric_vs_quality.png`).

TABLE **Table 5.4** — Spearman correlations of u_M , u_X , shared, coverage, the ratio and Q^2 against both the kinetic fit cost and the extrapolation performance.

CLAIM The ratio tracks out-of-domain performance at $\rho = +0.79$ to $+0.82$ ($p \approx 0.03$) and is the most stable of the candidate statistics across the two data generations. In-domain Q^2 reaches only $\rho = +0.32$ ($p = 0.48$).

CLAIM u_X alone is comparably predictive ($\rho = -0.86$ to -0.96) but has no scale; shared is strong on one data generation ($+0.96$) and fails on the other ($+0.43$). State this honestly — the ratio is chosen for stability and interpretability, not because it wins every single correlation.

CAUTION Add the sentence that the evaluation regime decides the answer: a single-shot evaluation gives the opposite sign to the rolling scheme, and only the rolling scheme tests what the metric is about.

5.9 Independent case study

Second process, two models defined by the source literature rather than constructed here.

VISUAL **Fig. 5.10** — three panels: each block alone, the commonality split, and both metrics on a log scale against the reference (`deck_cs2.png`).

TABLE **Table 5.5** — results table in the same format as Table 5.2 (`table_cs2_results.xlsx`).

CLAIM Breaking the reverse step collapses the mechanistic block from $Q^2 = 0.95$ to 0.59 while the measured block rises from 0.91 to 0.97. The correct model sits above the reference and the broken one below, on both responses, with $P(>1) = 0.00$ for the broken model in every replicate.

CAUTION Report $P(>1) = 0.95$ and 0.80 for the *correct* model and explain it rather than hiding it: u_x is $0.03 - 1.65\%$, so the magnitude is unstable even where the side of the threshold is not. This is the §2.7 degeneracy appearing exactly where predicted.

CLAIM Report the sample-size sensitivity: at 200 batches both responses reach $P(>1) = 1.00$ for the correct model and 0.00 for the broken one, and the relative spread falls to $0.25 - 0.29$. (`check_n_batches.xlsx`)

5.10 Scope conditions and limitations

Collect every limitation in one place, with evidence. A chapter that states its own limits precisely is harder to attack than one that leaves them to be found.

CLAIM Coverage is not a quality criterion. Across the candidates coverage correlates with the kinetic fit cost at $p = +0.93$ ($p = 0.003$) and with the ratio at $p = -0.82$ — the worst mechanism has the highest coverage. Selecting on both jointly would systematically choose worse models.

CLAIM Block layout matters. Truncating the measured block while the mechanistic block keeps the full horizon drives u_x to zero and can invert the ranking. Report the layout that was used and why.

CLAIM Insensitivity checks that passed. Giving the mechanistic block the full input programme rather than the truncated one changes no conclusion (`check_input_horizon.xlsx`).

CAUTION State the statistical limits: seven and two candidates, so rank correlations are indicative rather than established; and the second case study is a reconstruction.

Chapter 6 — Conclusions

Target length ~1,200 words.

6.1 Summary of the contribution

One paragraph per claim, in the order they were established: the ratio built on unique contributions has an intrinsic reference; it requires cross-validated Q^2 ; it separates sound from broken mechanisms on two independent processes; and it tracks out-of-domain performance where in-domain fit does not.

6.2 Practical recommendation

State the reporting rule compactly: report the ratio with coverage beside it, report the side of the reference rather than the magnitude when u_x is small, and state the block layout.

TO MAKE Table 6.1 — one-page practitioner summary: what to compute, what to report, when the metric does not apply.

6.3 Limitations

Short and direct; point to §5.10 rather than repeating it.

6.4 Future work

Obtaining the original input data for the second case study to convert the reconstruction into a replication; a defensible rule for when u_x is too small to divide by, based on its own cross-validation standard error; experimental design to reduce block redundancy and raise coverage; and a larger candidate family to test the threshold more severely.

Figures still to be produced: Fig. 2.1 (decomposition schematic), Table 2.1 (quantity definitions), Table 2.2 (applicability checklist), Fig. 3.1 (generic Måge schematic), Table 4.1 (candidate construction), Table 4.4 (design comparison), Table 6.1 (practitioner summary). Everything else named above already exists.